where m_n^* and m_p^* represent the effective masses of an electron and a hole respectively, E_g is the band gap, k is the Boltzmann's constant and T is the absolute temperature. Substituting n_i from Eq. (7.10) into (7.9), we get

$$\sigma = e(\mu_n + \mu_p) \frac{2(2\pi kT)^{3/2}}{h^3} \left(m_n^* m_p^*\right)^{3/4} \exp\left(-\frac{E_g}{2kT}\right) \tag{7.11}$$

which gives

$$\ln \sigma = -\left(\frac{E_g}{2k}\right) \frac{1}{T} + \frac{3}{2} \ln T + \text{constant} \tag{7.12}$$

The first term on the right hand side is the dominant term. The plot of $\ln\sigma$ versus $1/T$ is a straight line as shown in Fig. 7.5 The slope of the line gives an estimate of the band gap of the semiconductor.

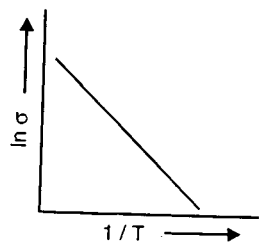


Fig. 7.5. *Plot of $\ln\sigma$ versus $1/T$.*

7.5 CARRIER CONCENTRATION AND FERMI LEVEL FOR INTRINSIC SEMICONDUCTOR

The concentration of electrons and holes in a semiconductor can be obtained from the knowledge of the densities of available states in the valence band and the conduction band as well as the Fermi-Dirac distribution function. The expression for the Fermi energy is then obtained from these carrier concentrations.

7.5.1 Electron Concentration in the Conduction Band

Referring to Eq. (5.44) the number of free electrons per unit volume in an energy range E and $E + dE$ can be written as

$$dn = D(E)\, f(E)\, dE \tag{7.13}$$

where $D(E)$ is the density of states defined as the total number of allowed electronic states per unit volume in a semiconductor and $f(E)$ is the Fermi distribution function representing the probability of occupation of a state with energy E. The expression for $f(E)$ is given by Eq. (5.28) as

$$f(E) = \frac{1}{\exp\left(\frac{E - E_F}{kT}\right) + 1} \tag{7.14}$$

whereas that for $D(E)$, which is strictly valid for free electrons, is obtained from Eq. (5.40) as

$$D(E) = \frac{4\pi}{h^3}(2m)^{3/2}E^{1/2} \tag{7.15}$$

Using Eqs. (7.15) and (7.13), we obtain

$$dn = \frac{4\pi}{h^3}(2m)^{3/2}E^{1/2}f(E)dE \tag{7.16}$$

It is apparent from Fig. 7.6 that an electron occupying an energy state E in the conduction band, in fact, possesses the kinetic energy $(E–E_c)$. Therefore, in Eq. (7.15), E must be replaced by $(E–E_c)$. Thus Eq. (7.16) becomes

$$dn = \frac{4\pi}{h^3}\left(2m_n^*\right)^{3/2}(E-E_c)^{1/2}\frac{1}{\exp\left(\dfrac{E-E_F}{kT}\right)+1}dE \tag{7.17}$$

where m_n^* is the effective mass of the electron in the conduction band. The concentration of electrons, n, in the conduction band is obtained by integrating Eq. (7.17) from $E = E_c$ to $E = \infty$, i.e.,

$$n = \frac{4\pi}{h^3}\left(2m_n^*\right)^{3/2}\int_{E_c}^{\infty}\frac{(E-E_c)^{1/2}dE}{\exp\left(\dfrac{E-E_F}{kT}\right)+1} \tag{7.18}$$

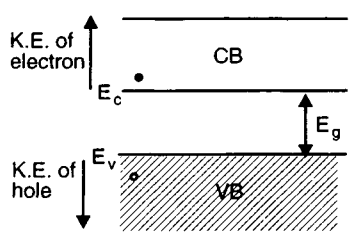


Fig. 7.6. *Band model of an intrinsic semiconductor.*

Now, near room temperature, $kT \cong 0.026$ eV. Therefore, for energies greater than E_c, we have

$$1 + \exp\left(\frac{E-E_F}{kT}\right) \cong \exp\left(\frac{E-E_F}{kT}\right)$$

$$\therefore\ n = \frac{4\pi}{h^3}\left(2m_n^*\right)^{3/2}\int_{E_c}^{\infty}(E-E_c)^{1/2}\exp\left[-\left(\frac{E-E_F}{kT}\right)\right]dE$$

$$= \frac{4\pi}{h^3}\left(2m_n^*\right)^{3/2}\exp\left(\frac{E_F-E_c}{kT}\right)\int_{E_c}^{\infty}(E-E_c)^{1/2}\exp\left[-\left(\frac{E-E_c}{kT}\right)\right]dE$$

Let $$\frac{E-E_c}{kT} = x$$

$\therefore$ $$dE = kT\,dx$$

For $E = E_c$, $x = 0$

$$\therefore \qquad n = \frac{4\pi}{h^3}\left(2m_n^*\right)^{3/2}\exp\left(\frac{E_F - E_c}{kT}\right)\int_0^\infty x^{1/2}(kT)^{1/2}e^{-x}kT\,dx$$

$$= \frac{4\pi}{h^3}\left(2m_n^* kT\right)^{3/2}\exp\left(\frac{E_F - E_c}{kT}\right)\int_0^\infty x^{1/2}e^{-x}\,dx$$

Now

$$\int_0^\infty x^{1/2}e^{-x}dx = \left(\frac{\pi}{4}\right)^{1/2}$$

$$\therefore \qquad n = 2\left(\frac{2\pi m_n^* kT}{h^2}\right)^{3/2}\exp\left[-\left(\frac{E_c - E_F}{kT}\right)\right] \tag{7.19}$$

From Eq. (7.14), the probability of occupancy of level E_c is given by

$$f(E_c) = \frac{1}{1+\exp\left(\dfrac{E_c - E_F}{kT}\right)} \cong \exp\left[-\left(\frac{E_c - E_F}{kT}\right)\right]$$

Therefore, Eq. (7.19) becomes

$$n = 2\left(\frac{2\pi m_n^* kT}{h^2}\right)^{3/2} f(E_c)$$

The first term on the right hand side must represent the effective density of states of electrons at the conduction band edge. Denoting it by N_c, we have

$$n = N_c \exp\left[-\left(\frac{E_c - E_F}{kT}\right)\right] \tag{7.20}$$

where

$$N_c = 2\left(\frac{2\pi m_n^* kT}{h^2}\right)^{3/2} \tag{7.21}$$

For silicon,